Data Structures (ETCCDS202)

Unit–1 Assignment: Foundations & Algorithmic Analysis

Course Details

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Part A: Stack ADT (ADT + Operations)

- What is **ADT**?
  - An **Abstract Data Type (ADT)** is a fundamental concept in computer science that essentially acts as a "contract" for how a collection of data should behave, without dictating *how* that behavior is actually implemented.
- Why stack operations are usually **O(1)** (constant time)?
  - Stack operations like push and pop are usually **O(1)** because we don't care about the size of the data—we only care about the top of the stack. In a stack, we don't search through the collection or shift elements around. We are always interacting with a single, known memory location.

Part B: Factorial & Fibonacci (Recursion + Complexity)

Factorial (n!)

For a simple iterative or recursive approach, the complexity is:

- **Time:** $\Theta(n)$ because it requires n multiplications.
- **Space:** $\Theta(n)$ for iterative; $\Theta(n)$ for recursive due to the call stack.

  Big- $\Omega(n)$ and Big-O(n) match here, making it tightly bound by $\Theta(n)$.

Fibonacci (F_n)

The complexity for the **naive recursive** approach is:

- **Time:** $O(2^n)$
- **Space:** $\Theta(n)$ due to the maximum depth of the recursion tree.

  Iterative versions or dynamic programming reduce time to $\Theta(n)$ and space to $\Theta(1)$.

Why Naive Fibonacci is Slow?

The naive recursion performs **redundant calculations**. To find F_n, it recalculates the same subproblems (like F_{n-2}) thousands of times across different branches. This results in an **exponential growth** of function calls, making it computationally expensive and inefficient for even modest values of n.

Part C: Tower of Hanoi (Recursion + Trace)

Why the number of moves is $2^n – 1$?

The recurrence relation for n disks is $T(n) = 2T(n-1) + 1$. To move n disks, you must move the top n-1 disks twice (once to the auxiliary and once to the destination) plus

one move for the largest disk. Solving this geometric series results in exactly $2^n - 1$ moves.

Time Complexity: $O(2^n)$

The algorithm generates a binary tree of recursive calls. Each disk added doubles the number of operations required, as every call for n disks triggers two subsequent calls for n-1 disks. This exponential growth means the execution time increases significantly with each additional disk, following an $O(2^n)$ pattern.

Space Complexity: O(n)

The space complexity is determined by the maximum depth of the recursion stack. Since the algorithm processes one branch at a time and the recursion depth corresponds directly to the number of disks n, the system only needs to store n stack frames at any given moment, resulting in O(n) space.

Part D: Recursive Binary Search (Divide & Conquer + Recurrence)

Time Complexity and Recurrence

The recurrence $T(n) = T(n/2) + O(1)$ represents dividing the search space in half each step and performing a constant-time comparison. By repeatedly halving the input size, the number of steps required to reach a single element is $\log_2 (n)$, leading to a complexity of O(log n).

Best, Average, and Worst Cases

- **Best Case:** Occurs when the target key is exactly at the middle index of the initial array, requiring only one comparison: O(1).
- **Average Case:** Occurs when the key is found after several divisions or is absent, typically involving log(n) comparisons: O(log n).
- **Worst Case:** Occurs when the key is at the far ends of the array or not present at all, requiring the maximum number of divisions: O(log n).

Part E: Short Paradigm Reflection

Binary Search as Divide & Conquer

Binary Search is a **Divide & Conquer** algorithm because it recursively breaks the search space into smaller halves. It **divides** the array by finding the midpoint, **conquers** by comparing the key to that midpoint, and reduces the problem to a single sub-array, ignoring the irrelevant half.

Fibonacci and Memoization

Memoization transforms the recursive Fibonacci calculation into **Dynamic Programming** by storing the results of expensive function calls. It solves the **overlapping subproblems** issue, preventing the algorithm from redundantly recalculating the same Fibonacci numbers multiple times, which reduces time complexity from exponential to linear.

Tower of Hanoi Recursion

The **base case** occurs when n = 1, where the disk is moved directly to the destination. The **smaller subproblem** involves moving n-1 disks to an auxiliary peg to clear a path for the largest disk, then moving those n-1 disks from the auxiliary to the final destination.

Real-Life Examples

- **Divide & Conquer:** A dictionary search. You open to the middle, determine if your word is before or after that page, and repeat the process on the remaining half.
- **Dynamic Programming:** Google Maps navigation. It breaks the trip into segments and reuses previously calculated shortest paths for common intersections to find the overall fastest route.
- **Greedy:** A cashier giving change. They always choose the largest denomination possible first (e.g., a $20 bill) to minimize the total number of bills/coins handed back.